

Grundlagen der künstlichen Intelligenz

Los Del DGIIM, losdeldgiim.github.io

Doble Grado en Ingeniería Informática y Matemáticas
Universidad de Granada

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1. Introduction

Artificial Intelligence (AI) is a field that is rapidly evolving and has the potential to revolutionize various industries. Before delving into the technical aspects of AI, it is important to understand the concept of intelligence and how it can be measured.

1.1. Intelligence Tests

Scientists have tried to decide what differentiates AI from computer algorithms; i.e. what is the essence of intelligence? This question has been debated for decades, and there are different tests that have been proposed to determine whether a machine can be considered intelligent or not. The importance of these tests is shown in the Chinese Room thought experiment.

- Chinese Room Experiment:

The Chinese Room thought experiment involves a person who does not understand Chinese sitting in a room with a set of instructions for how to respond to Chinese characters. The person receives Chinese characters as input and uses the instructions to produce appropriate responses, which are then sent back out of the room.

From the outside, it may appear that the person in the room understands Chinese, but in reality, they are simply following a set of rules without any comprehension of the language. This thought experiment raises questions about whether a machine can truly understand language or if it is simply following a set of instructions.

Some of the most well-known tests include:

- Turing Test:

In the Turing Test, a human evaluator interacts with both a machine and a human without knowing which is which. If the evaluator cannot reliably distinguish between the machine and the human, then the machine is said to have passed the Turing Test and is considered to be intelligent.

It only tests for the ability to mimic human behavior, not for true understanding or consciousness.

- Winograd Test:

The test involves a series of sentences known as “Winograd Schemas”. They include ambiguous pronouns that require common sense and contextual understanding to resolve.

This test is not about factual knowledge but rather about understanding context.

- Lovelace Test:

The Lovelace Test is designed to evaluate a machine's ability to create something original and surprising.

This test is extremely difficult for machines to pass, as they often just combine existing ideas rather than creating something truly new.

- Metzinger Test:

The Metzinger Test is designed to evaluate a machine's ability to have a subjective experience, a sense of “self, I”, and consciousness. There is no definitive method for conducting the test in practice, since subjective experience cannot be observed from the outside.

1.2. Initial Definitions

From more general concepts to more specific ones, where each one builds upon the previous, some important terms in the field of AI include:

- Data Science: Field that turns data into knowledge and decisions.
- Artificial Intelligence: Field that aims to create machines that can perform tasks that typically require human intelligence.
- Machine Learning: Subfield of AI that focuses on training machines to learn from data and improve their performance over time without being explicitly programmed. There are three main types of machine learning:
 - Supervised Learning: The machine is trained on labeled data, where the correct output is provided for each input.
 - Unsupervised Learning: The machine is trained on unlabeled data, where it must find patterns and relationships on its own.
 - Reinforcement Learning: The machine learns by interacting with an environment and receiving feedback in the form of rewards or penalties based on its actions.
- Neural Networks: A type of (usually supervised) machine learning model inspired by the structure and function of the human brain. They consist of layers of interconnected nodes (neurons) that process and transmit information.
- Deep Learning: Neural networks with multiple layers. Deep learning has been particularly successful in tasks such as image recognition, natural language processing, and speech recognition.

2. Data Science

Data Science is a multidisciplinary field that combines statistical analysis, machine learning, and domain expertise to extract insights and knowledge from data. Data is the raw material that fuels the development of AI systems, and understanding how to work with data is crucial for anyone interested in the field of AI. Two main people work on it:

- Data Engineer: Responsible for building and maintaining the infrastructure that allows for the collection, storage, and processing of data.
- Data Scientist: Responsible for analyzing and interpreting data to extract insights and knowledge.

Regarding data collection, there are some aspects that must be taken into account:

- Data Quality: The quality of the data is crucial for the success of any AI project.
- Data Integrity: Ensuring that the data is accurate and reliable.
- Data Consistency: Ensuring that the data is consistent across different sources and formats.

Once the data has been collected, it must be preprocessed before it can be used for training AI models. This involves:

- Data Validation: Checking the data for errors and inconsistencies. Depending on the source and type of data, some rules can be applied to validate it.
- Data Cleaning: Removing or correcting any errors or inconsistencies in the data.
- Data Transformation: Converting the data into a format that can be used for training AI models.

Data preprocessing also involves data scaling, which is the process of normalizing or standardizing the data to ensure that it is on the same scale. This is important because many AI algorithms are sensitive to the scale of the data and can perform poorly if the data is not scaled properly. Some common scalers include:

- **Binarizer**: Converts categorical data into binary data. A threshold is set, and values above the threshold are converted to 1, while values below the threshold are converted to 0.

$$X' = \begin{cases} 0 & \text{if } X \leq \text{threshold} \\ 1 & \text{if } X > \text{threshold} \end{cases}$$

- **MaxAbsScaler**: Scales the data to the range $[-1, 1]$ by dividing each value by the maximum absolute value in the data.

$$X' = \frac{X}{\max(|X|)}$$

- **MinMaxScaler**: Scales the data to a specified range, let's say $[\text{mín}, \text{máx}]$. Firstly the data is scaled to the range $[0, 1]$ (X_{std}), and then it is scaled to the specified range.

$$X_{\text{std}} = \frac{X - \text{mín}(X)}{\text{máx}(X) - \text{mín}(X)} \quad X' = X_{\text{std}} \cdot (\text{máx} - \text{mín}) + \text{mín}$$

- **Normalizer**: Scales the data to have a unit norm. This is done by dividing each value by the norm of the data. Usually, the L2 norm is used.

$$X' = \frac{X}{\|X\|}$$

- **PowerTransformer**: Applies a power transformation to the data to make it more Gaussian-like. This is useful for data that is not normally distributed.
- **RobustScaler**: Scales the data using statistics that are robust to outliers. This is useful for data that contains outliers.

$$X' = \frac{X - \text{median}(X)}{\text{IQR}(X)}$$

where $\text{IQR}(X)$ is the interquartile range of X .

- **StandardScaler**: Scales the data to have a mean of 0 and a standard deviation of 1.

$$X' = \frac{X - \mu}{\sigma}$$

where μ is the mean of X and σ is the standard deviation of X .

One important aspect of data preprocessing is *encoding* the data. This is the process of converting categorical data into numerical data that can be used for training AI models. There are several techniques for encoding data, such as:

- **Ordinal Encoding**: Assigning a unique integer to each category in the data. The order of the integers is important, as it implies a relationship between the categories.

$$\{\text{small, medium, large}\} \rightarrow \{0, 1, 2\}$$

- Label Encoding: Similar to ordinal encoding, but it does not imply any relationship between the categories. Each category is assigned a unique integer, but the order of the integers does not matter.

$$\{\text{red, green, blue}\} \rightarrow \{0, 1, 2\}$$

- One-Hot Encoding: Creating a binary column for each category in the data.

$$\{\text{red, green, blue}\} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Dummy Encoding (Leave out the last): Similar to one-hot encoding, but it leaves out the last category to avoid multicollinearity.

$$\{\text{red, green, blue}\} \rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

- Absolute Frequency Encoding: Assigning a unique integer to each category based on the absolute frequency of the category in the data.

$$\begin{aligned} &\{\text{red, red, red, green, blue, blue}\} \\ &\{\text{red, green, blue}\} \rightarrow \{3, 1, 2\} \end{aligned}$$

- Relative Frequency Encoding: Assigning a unique integer to each category based on the relative frequency of the category in the data.

$$\begin{aligned} &\{\text{red, red, red, green, blue, blue}\} \\ &\{\text{red, green, blue}\} \rightarrow \{3/6, 1/6, 2/6\} \end{aligned}$$

- Feature Hash Encoding: Hashing the categories into a fixed number of columns. This is useful for data with a large number of categories.

Once the data has been preprocessed, they can be used. There are two main destinations for the data:

- Data Visualization: The process of creating visual representations of data to help understand and communicate insights. This is used by people.
- Data Modeling: The process of using data to train AI models. This is used by machines.

3. Search Algorithms

3.1. Sorting Algorithms

3.2. Uninformed Search

3.3. Informed Search

3.3.1. Dijkstra Search

3.4. Local Search

3.5. Modern utilities

3.5.1. Information Retrieval

3.5.2. Search Engines

3.5.3. PageRank

4. Übungen

4.0. Intro to Python

The Python part of this exercise is available in [this Jupyter Notebook](#).